LEROY MATIENGA

Computer Science Undergraduate & Software Developer

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PROFESSIONAL SUMMARY

Motivated and detail-oriented Computer Science undergraduate with strong foundations in software engineering, backend systems, and interactive web graphics. Experienced in building full-stack web applications with Python Flask and SQLite, engineering real-time 3D WebGL visualizations with Three.js, and developing AI-powered software solutions.

TECHNICAL SKILLS

Programming Languages:	Python (OOP, Data Structures), JavaScript (ES6+), SQL, HTML5, CSS3
Frameworks & 3D:	Flask, Three.js (WebGL, Custom GLSL Shaders), OrbitControls, Canvas API
Databases & Tools:	SQLite, Git, GitHub, VS Code, REST APIs, JSON, smtplib
Key Competencies:	Backend API Design, Input Sanitization (XSS Defense), Rate Limiting, 3D Coordinate Mapping

FEATURED PROJECTS

Interactive 3D Real-Time Earth Portfolio | *Three.js, GLSL, Python Flask, SQLite*

- Developed a hardware-accelerated 3D WebGL Earth globe featuring astronomical solar vector tracking based on live UTC time.
- Authored custom GLSL vertex and fragment shaders computing real-time day/night terminator blending, twilight Rayleigh scattering, and night city lights.
- Engineered a full-stack Flask REST API with sliding-window rate limiting, XSS input sanitization, directory traversal guards, and SQLite persistence.

MUSA Assistant (AI & Academic Management System) | *Python, NLP, REST APIs*

- Engineered an intelligent voice- and text-activated academic assistant designed to streamline university student workflows.
- Integrated NLP queries for automated timetable tracking, exam schedule alerts, and institutional portal notifications.

Enterprise Inventory Management GUI | *Python, Tkinter, SQLite*

- Designed a desktop inventory control application applying strict Object-Oriented Programming (OOP) design patterns.
- Implemented ACID-compliant atomic transactions, fast search indexing, and automated PDF/CSV business report generators.

Dynamic Data Visualizer | *JavaScript, HTML5 Canvas, Algorithms*

- Built an interactive web analytics platform providing real-time data charting, algorithm step-by-step sorting visualizations, and CSV parsing.

EDUCATION

Bachelor of Science in Computer Science (Undergraduate)

Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems Architecture, Operating Systems, Computer Networks, Software Engineering Principles.